Mock 6-10: Error Notes

CFA Level II · 59 notes (45 wrong, 14 right but unsure) · grouped by subject · mistake → fix, with the formula for calculation questions

How to read: red border = wrong answer (your pick → correct answer); yellow border = right but unsure. Each note gives the mistake, the fix, and a formula box where the question was a calculation. Count by mock: Mock 10 10, Mock 6 10, Mock 7 11, Mock 8 15, Mock 9 13. Mock 8 notes follow the version you sat.

Ethics (3)

Mock 6 Q43 · you A → answer C

Mistake: In a 'least likely violated' question I picked a Standard that WAS violated (Communication with Clients).

Fix: Mark every option violated / not violated first, then pick the one not violated. Here Fair Dealing was the one not breached; V(B) (fact vs opinion, disclose the process) and VI(B) (clients before own trades) were.

Mock 8 Q39 · you C → answer A

Mistake: Treated 'disclose the finder's fee in the prospectus' as the wrong action.

Fix: VI(C) Referral Fees: disclose to employer, clients and prospects BEFORE the service, and the recommended update is at least QUARTERLY, not semiannually. Disclosure in the prospectus is fine.

Mock 10 Q9 · you A → answer B

Mistake: Thought paid travel from a company I cover is acceptable when it does not change my decision.

Fix: I(B) Independence and Objectivity: always use commercial transport at your own or your firm's expense. A token gift is fine; paid travel is the violation (the appearance of compromise counts). Exception: remote site with no commercial transport.

FSA (8)

Mock 6 Q23 · you C → answer B

Mistake: Picked 'low fluctuation of the cash-flow accruals ratio' as a sign of poor performance quality.

Fix: Red flags are: accruals ratio trending UP or erratic, and operating income consistently ABOVE cash from operations (Reason 3). Steady or falling accruals are not a concern.

```
Balance-sheet accruals = (NOA end − NOA begin) ÷ average NOA
Cash-flow accruals = (NI − CFO − CFI) ÷ average NOA
```

Mock 8 Q30 · you B → answer C

Mistake: Said a pension discount rate above peers is conservative.

Fix: Higher discount rate → lower PV of the obligation and lower expense → AGGRESSIVE (deficit understated). Lower rate = conservative.

```
PBO = Σ benefit_t ÷ (1 + discount rate)^t: rate ↑ → PBO ↓
```

Mock 8 Q31 · you A (right) unsure

Unsure: Right, but unsure why both quality labels apply.

Fix: Consistent losses on asset sales mean depreciation was understated in earlier years: operating income was overstated (low earnings quality) and the bias was deliberate and lagged (low financial reporting quality).

Mock 8 Q32 · you C → answer B

Mistake: Said the debt-to-equity ratio decreases after cutting the pension discount rate by 0.5%.

Fix: Lower discount rate → larger obligation → liabilities ↑, equity ↓ (via AOCI), assets unchanged. So D/E ↑, financial leverage ↑ and book value per share ↓.

```
D/E = total debt ÷ equity: numerator ↑, denominator ↓ → ↑
```

Mock 9 Q25 · you C → answer A

Mistake: Linked Basel III to earnings quality.

Fix: Basel III raised the equity funding required against risk-weighted assets (limits leverage) and favours stable funding: consumer deposits are more stable than interbank. Earnings quality was not its target.

Mock 9 Q26 · you C → answer A

Mistake: Expected underwriting standards to tighten after a hard market.

Fix: Hard market: high premiums and profits attract new competitors, the market softens, underwriting standards fall and capital is depleted (capital adequacy gets harder).

Mock 9 Q28 · you B → answer C

Mistake: Looked only at the loss ratio below 100%.

Fix: Combined ratio adds the loss ratio and the underwriting expense ratio. Above 100% = underwriting loss that investment income must cover.

```
Combined ratio = (claims + loss adj. expenses) ÷ net premiums earned + underwriting expenses ÷ net premiums written
```

Mock 10 Q1 · you C → answer B

Mistake: Read the DuPont trend in the wrong direction (columns were newest year first).

Fix: Check the column order of the years before judging a trend. Rebuild assets and equity first, then compare leverage across years.

```
Total assets = revenue ÷ asset turnover
Equity = assets − liabilities
Financial leverage = assets ÷ equity
```

Equity (8)

Mock 6 Q25 · you B → answer A

Mistake: Picked R65m: MVA must use the market value of the WHOLE firm (debt and equity) against total book capital.

Fix: MVA = market value of the firm − book value of total capital (market value of the firm includes the debt).

```
MVA = (MV debt + shares × price) − total book capital
= 55 + 30 × 25.43 − 650 = R168m
```

Mock 6 Q26 · you A → answer B

Mistake: Said residual income is zero although intrinsic value (R22) is above book value (R20).

Fix: V0 > B0 → PV of future RI is positive; V0 = B0 → zero; V0 < B0 → negative.

```
V0 = B0 + PV(residual income)
```

Mock 7 Q12 · you B → answer C

Mistake: Multistage residual income value came out at €83.01 instead of €85.71 (the continuing value of RI was not built right).

Fix: Build RI year by year, then add the value of RI after the last year. Persistence factor 1 means RI stays constant forever.

```
RI_t = EPS_t − r × B_{t−1}
V0 = B0 + Σ RI_t ÷ (1 + r)^t + PV(continuing RI)
Continuing RI at T = RI_T × ω ÷ (1 + r − ω)
```

Mock 7 Q38 · you B → answer C

Mistake: Believed a high P/E is also justified by high risk.

Fix: High growth justifies a higher P/E, but higher risk raises the required return and justifies a LOWER P/E.

```
Justified P/E = payout ÷ (r − g): g ↑ → P/E ↑, r ↑ → P/E ↓
```

Mock 7 Q39 · you B → answer A

Mistake: Used the simple average (13.8) for an index P/E.

Fix: Index P/E uses the weighted HARMONIC mean (average the earnings yields). Order: harmonic < geometric < arithmetic.

```
Weighted harmonic = Σ weights ÷ Σ(weight ÷ P/E) = total market cap ÷ Σ(cap ÷ P/E) = 11.3
```

Mock 8 Q20 · you B (right) unsure

Unsure: Right, but unsure when the H-model stops working.

Fix: H-model fits a LINEAR fade from high to low growth; use a spreadsheet multistage model for a long high-growth period or a large gap between growth rates.

```
V0 = D0(1 + gL) ÷ (r − gL) + D0 × H × (gS − gL) ÷ (r − gL),   H = half-life = t ÷ 2
```

Mock 8 Q25 · you B (right) unsure

Unsure: Right, but unsure how to choose the model.

Fix: FCFF when assets are largely intangible and dividends are only occasional or special. DDM needs a consistent dividend; asset-based needs marketable assets.

Mock 9 Q15 · you A → answer B

Mistake: Chose P/B for a company whose value is its brand.

Fix: P/B ignores internally built intangibles. High leverage and mixed capital structures → EV/sales. Dividend yield ignores capital gains.

```
EV = market cap + debt + preferred − cash
```

Fixed Income (7)

Mock 6 Q40 · you B → answer C

Mistake: Said log-normality is not an input of tree calibration.

Fix: Calibration uses a volatility assumption, an interest rate model (log-normal) and the market values of the benchmark bonds. The bond being valued is NOT an input (it is valued afterwards on the tree).

Mock 7 Q30 · you C → answer A

Mistake: Added the migration effect to the yield (sign flipped).

Fix: Expected return = YTM + expected % price change. A spread widening lowers the price.

```
E(R) = YTM + Σ prob_i × [−MD × (spread_i − spread_0)]
= 5.50% − 1.77% = 3.73%
```

Mock 7 Q31 · you B → answer C

Mistake: Got 0.9388% instead of 1.4082%: one step of the CVA chain was off.

Fix: Always do three steps: VND from the tree, then CVA, then the yield of (VND − CVA) minus the benchmark yield.

```
CVA = Σ DF(benchmark) × PD × LGD × expected exposure
Fair value = VND − CVA
Spread = YTM(fair value) − benchmark YTM
```

Mock 8 Q10 · you A → answer B

Mistake: Used the wrong loss given default (0.7796% is what LGD 50% gives).

Fix: Recovery of one-quarter means LGD = 75%. Convert recovery to LGD before anything else.

```
LGD = 1 − recovery rate
PD_t = h × (1 − h)^(t−1)
CVA = Σ DF × PD × LGD × exposure
```

Mock 8 Q11 · you C (right) unsure

Unsure: Right, but unsure of the direction.

Fix: Credit spread falls when the risk-neutral PD falls (CVA ↓ → fair value ↑ → yield ↓). A higher PD or higher LGD widens the spread.

Mock 9 Q6 · you B → answer C

Mistake: Confused liquidity preference with pure expectations.

Fix: Pure expectations: forward rates are unbiased predictors of future spot rates (the future curve = today's forward curve). Liquidity preference: forward = expected spot + liquidity premium. Segmented markets: each maturity has its own supply and demand.

Mock 9 Q8 · you A (right) unsure

Unsure: Right, but unsure how the two views combine.

Fix: Flight to quality → yields fall → extend duration. Bullish flattening (long rates fall more than short) → barbell rather than bullet.

Derivatives (5)

Mock 6 Q36 · you A → answer B

Mistake: Accepted Jeffries' claim that the smile and the skew take the same shape.

Fix: Implied volatility is backed out of market option prices and is forward-looking. The volatility surface shows IV across strikes AND maturities. Smile and skew do not have the same shape (hedging demand steepens the skew).

Mock 7 Q23 · you C → answer B

Mistake: Took the fixed rate of the wrong currency (the Korean rate, which the Foundation pays).

Fix: Decide which leg is received and which is paid by currency first. Compute each fixed rate from its own country's discount factors.

```
Swap fixed rate = (1 − PV_n) ÷ Σ PV_i  (that country's factors)
= (1 − 0.9260) ÷ 4.7995 = 1.54%
```

Mock 8 Q1 · you C → answer B

Mistake: Left the bond cost uncompounded in the bond futures arbitrage.

Fix: Carry the bond price plus accrued interest forward to expiry. Futures side = CF × futures price + accrued interest at expiry. Discount only the final difference.

```
F0 = (1 + r)^T × (B0 + AI0)
Implied = CF × Q0 + AI_T
Profit = (implied − F0) ÷ (1 + r)^T
```

Mock 10 Q23 · you B (right) unsure

Unsure: Right, but unsure which Greek to use.

Fix: 'All else equal, one day later' means only time decay: value falls by theta.

```
New value = old value + theta × days  (theta is negative)
Full: ΔC ≈ Δ·ΔS + ½Γ·ΔS² + Θ·Δt
```

Mock 10 Q24 · you A → answer B

Mistake: Gave the European put value (8.45); forgot early exercise at the down node.

Fix: American option: at EVERY node take the larger of the binomial value and the exercise value. Down node: 20 > 16.77, so use 20.

```
π = (1 + r − d) ÷ (u − d)
V = [π V_u + (1 − π) V_d] ÷ (1 + r)
American: V = max(V, K − S)
```

Quant (7)

Mock 7 Q33 · you B → answer A

Mistake: Called the variable significant although |t| was below the critical value.

Fix: Compare |t| with the critical value for n − k − 1 degrees of freedom: STIM |−2.10| < 2.12 (not significant), CRISIS 2.35 > 2.12 (significant).

```
t = coefficient ÷ standard error
```

Mock 8 Q15 · you C (right) unsure

Unsure: Right, but unsure how ARCH is tested.

Fix: Regress SQUARED residuals on their own lag: t above the critical value → ARCH. Residuals on their own lag test serial correlation.

```
ε²_t = a0 + a1 ε²_{t−1}
```

Mock 9 Q34 · you C → answer A

Mistake: Accepted that SEE measures errors of an independent variable.

Fix: SEE is the standard deviation of the differences between actual and estimated values of the DEPENDENT variable. Small SEE means strong fit.

```
SEE = √[SSE ÷ (n − k − 1)]
```

Mock 9 Q36 · you B (right) unsure

Unsure: Right, but unsure which test applies.

Fix: One slope coefficient: t-test. All slopes together: F-test. R^2 and SEE describe the fit, not a single coefficient.

Mock 10 Q13 · you C → answer A

Mistake: Expected the slope of a random walk to be near 0.

Fix: Random walk as an AR(1): b0 ≈ 0 and b1 ≈ 1 (unit root).

```
x_t = b0 + b1 x_{t−1} + ε:  b0 ≈ 0, b1 ≈ 1
```

Mock 10 Q14 · you A (right) unsure

Unsure: Right, but unsure.

Fix: For a random walk the best forecast of the next value is the CURRENT value (no mean reversion, not covariance stationary).

Mock 10 Q15 · you C (right) unsure

Unsure: Right, but unsure.

Fix: ARCH(1) is present when the slope c1 on the lagged squared residual is significantly different from 0.

Portfolio Management (6)

Mock 6 Q31 · you A → answer B

Mistake: Called filling a temporary benchmark gap 'rebalancing'.

Fix: ETF uses: completion = fill a gap in benchmark exposure; rebalancing = return to target weights after market moves; liquidity management = equitize cash for inflows and outflows.

Mock 7 Q41 · you C → answer A

Mistake: Picked 6.00% instead of the model intercept (2.58%).

Fix: In a macro factor model the factors are SURPRISES; expected return is the intercept when all surprises are zero.

```
R_i = a_i + b_i1 F1 + b_i2 F2 + … + ε;
E(R_i) = a_i
```

Mock 7 Q43 · you A → answer C

Mistake: Judged one statement wrong although both were true.

Fix: Macro model: factors = surprises in economic variables, sensitivities estimated LAST. Fundamental model: factors = company attributes, sensitivities specified FIRST.

Mock 7 Q44 · you A → answer C

Mistake: Agreed that multifactor models are useless for passive management.

Fix: Multifactor models serve BOTH active management (tilts, risk) and passive management (replicate an index's factor exposures).

Mock 9 Q41 · you B → answer C

Mistake: Called VaR 'subjective' as an advantage.

Fix: VaR advantages: basis for risk-adjusted performance evaluation, comparable across portfolios. Its limits: not a worst case, subjective inputs.

Mock 9 Q43 · you A → answer B

Mistake: Described historical simulation, not the parametric method.

Fix: Parametric VaR assumes normally distributed returns: poor for fat tails and options. Historical uses past data; Monte Carlo needs many random scenarios.

```
VaR = (μ − z × σ) × portfolio value
```

Economics (4)

Mock 6 Q19 · you A → answer C

Mistake: Counted a low exchange rate as a growth channel on the supply side.

Fix: Capital deepening and investment in technology raise potential growth. A low currency supports export demand and has little direct effect on supply capacity.

Mock 7 Q20 · you B → answer C

Mistake: Chose a central bank action that expands money when inflation is the threat.

Fix: Inflation threat → contract money: sell domestic securities (open market), raise rates.

Mock 8 Q6 · you A → answer C

Mistake: Accepted a false inflation-linked bond statement.

Fix: Break-even inflation = nominal yield − real yield (same maturity). More inflation uncertainty → HIGHER break-even (inflation risk premium). It does not compare long and short linkers.

```
BEI = y(nominal) − y(inflation-linked)
```

Mock 8 Q8 · you C → answer B

Mistake: Misread credit spreads over the business cycle.

Fix: In a downturn spreads widen and high-rated bonds outperform. When spreads narrow, weaker credits improve MORE.

Corporate Issuers (4)

Mock 6 Q16 · you B → answer A

Mistake: Said ESG integration lowers the credit spread for a company with poor governance.

Fix: Few independent directors, weak diversity and high social risk → higher risk perception → credit spread INCREASES.

Mock 8 Q34 · you C → answer A

Mistake: Compared the companies without scoring each feature.

Fix: Score every feature separately (1 point per good-governance feature), then add: here the totals were the same.

Mock 9 Q9 · you B → answer A

Mistake: In an 'INCORRECT statement' question I picked a correct statement.

Fix: Retiring debt improves the rating and LOWERS the cost of capital. Sale-leasebacks unlock real estate value. Re-read whether the question asks for correct or incorrect.

Mock 9 Q12 · you A (right) unsure

Unsure: Right, but unsure.

Fix: Spin-off: shares of the subsidiary go to existing shareholders, no cash. Equity carve-out: sells shares to new investors, cash to the parent. Divestiture: sells the segment.

Alternative Investments (7)

Mock 6 Q2 · you C → answer A

Mistake: Thought a REIT needs less expertise than a REOC.

Fix: REIT is a pass-through (no corporate tax, no double taxation). A REOC is taxed as a company and can hold properties under development.

Mock 8 Q43 · you B (right) unsure

Unsure: Right, but unsure how the merger arbitrage gain works.

Fix: Long target, short acquirer. If the deal closes, the target shares convert into acquirer shares that cover the short.

```
Gain = short proceeds − cost of the target
shares
= 350,000 − 330,000 = $20,000
```

Mock 8 Q44 · you B (right) unsure

Unsure: Right, but unsure of the main risk.

Fix: The main risk is deal failure: the target falls back to its pre-announcement price while the short position no longer offsets it.

Mock 9 Q40 · you B → answer A

Mistake: Matched obsolescence of space to a storage REIT.

Fix: Storage REIT → new competing facilities (easy entry). Office REIT → obsolescence. Retail REIT → tenants' sales per square foot.

Mock 10 Q37 · you A → answer B

Mistake: Compared dollar amounts of unreimbursed expenses.

Fix: Compare unreimbursed expenses as a % of effective gross income: Boston 44.5% vs New York 36.7%, so Boston is hit harder.

```
Unreimbursed expenses % = (operating
expenses − reimbursements) ÷ EGI
```

Mock 10 Q38 · you A → answer C

Mistake: Thought both statements could describe the same cycle phase.

Fix: High vacancy = oversupply or recession phase, so only a recovery within two years (Statement 2) fits. The two statements imply different phases.

Mock 10 Q40 · you A (right) unsure

Unsure: Right, but unsure.

Fix: Direct capitalisation uses one period's NOI. If NOI is temporarily disrupted, use a NORMALIZED NOI. Sales comparison is less suited while a recovery is expected.

```
Value = NOI ÷ cap rate
```